DAY 05 • STUDENT HANDOUT

Checking for Alerts

Receiving detection alerts and incident reports, and checking for red-flag detections as per SOP

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Element 1 Check for alerts: Performance Criteria 1.1–1.3; Range of Variables: Detection alert, Detection report, Red flag detection; Required Knowledge 1.5 Sources of detection alert / incident reports, 1.6 Log and detection management, 1.7 Security solution scan and operations procedure, 1.10 Malicious software behaviors
Duration	8 hours (1 training day)

Learning Objectives

At the end of this session, you should be able to:

1. Explain what a Standard Operating Procedure (SOP) is and why alert intake follows one.
2. Receive and record detection alerts from SIEM, AV, firewall, WAF, DLP, EDR and NDR as per SOP (PC 1.1).
3. Receive and record incident reports made by phone, walk-in, email, SMS, chat and video conference (PC 1.2).
4. Check detections for ransomware and PE infection (virus) red flags as per SOP (PC 1.3).
5. Recognize social engineering attempts during intake and respond as per SOP.

1. What "Checking for Alerts" Means

Checking for alerts is the first element of monitoring and reporting cyber threats. A monitoring analyst receives every alert and every report, records it completely, and checks it at once for red flags. Each organization writes these steps in a Standard Operating Procedure (SOP), so that every analyst on every shift does the same thing, in the same order, within the same time limits. In the lab you follow SOC-SOP-01 Alert and Report Intake (~/day5/soc_alert_sop.txt).

1. Receive — take the detection alert from the queue, or the report from a person (PC 1.1, 1.2).
2. Record — write down the facts the SOP asks for, without judging yet.
3. Check red flags — compare what you see with the SOP red flag list (PC 1.3). Red flags are escalated immediately.
4. Hand over — assessment against pre-identified criteria and ticketing follow on Day 6 (PC 1.4, 1.5).

2. Receiving Detection Alerts (PC 1.1)

A detection alert is a message from a security solution saying it saw something suspicious. The Competency Standards list seven sources. They all arrive in the SOC alert queue, usually through the SIEM.

Detection Alert (CS)	What It Reports	Example from the Day 5 Queue
SIEM alert	Correlated events from many logs	"18 logon failures then success on DC-01"

Detection Alert (CS)	What It Reports	Example from the Day 5 Queue
AV alert	Malware found on an endpoint	"Trojan:Win32/Wacatac.B!ml quarantined"
Firewall alert	Traffic allowed or denied by rules	"Inbound RDP denied (64 attempts in 5 min)"
Web application firewall (WAF) alert	Attacks against websites	"SQL injection blocked on /loan/apply.php"
DLP alert	Sensitive data leaving the organization	"Payroll file uploaded to personal cloud storage"
EDR alert	Suspicious behavior on an endpoint	"Mass file rename by unsigned process"
NDR alert	Suspicious network behavior	"Unusual SMB writes to \\FS-01\Installers"

Acknowledge on Time

To acknowledge an alert means to take ownership of it: you change its status from New to Acknowledged so that everyone knows someone is working on it. The SOP sets a time limit for each severity. Missing it is an SLA breach (SLA = service level agreement).

Severity	Acknowledge Within (SOC-SOP-01 rule A2)
Critical	15 minutes
High	30 minutes
Medium	120 minutes
Low	480 minutes (8 hours)
Informational	No acknowledgment needed (auto-closed)

> **TIP**
> Waiting time = now − received time. Alert Q-5006 (High) was received at 08:33:40. At 09:30 it has waited 56 minutes, which is more than 30, so it is an SLA breach: acknowledge it at once and tell the Shift Lead. Always use the same time zone (Philippine Standard Time, UTC+8).

What to Record for Every Alert

- Alert ID, time received, and time acknowledged
- Source (SIEM, AV, firewall, WAF, DLP, EDR, NDR) and product
- Host (affected computer) and rule or detection name
- Severity and status
- Whether it matches a red flag

3. Receiving Incident Reports from People (PC 1.2)

Not every threat is found by a tool. Employees, customers and partners also report what they see. The Competency Standards list six channels for a detection report:

Channel (CS)	Example	Intake Tip
Phone call	An accountant calls: "my files end in .lockd"	Write their exact words; confirm the callback number
Walk-in	An engineer comes to the SOC desk	Ask what they did and when; note the PC name
Email	A branch forwards a suspicious email	Ask them to forward it as an attachment; do not click links
SMS	"IT" asks the SOC to text an OTP	Treat requests for passwords or OTPs as social engineering
Chat	A user reports a printer jam	Not a security matter: redirect to the IT Helpdesk
Video conference	A fake "virus warning" pop-up is shared on screen	Record what you saw; tell them not to call the number

The Intake Record

- Report ID, channel, date and time received
- Reporter name, department and callback number from the company directory
- What they saw, in their own words
- Affected computer, account or device
- What they already did (clicked, opened, restarted)

> **NOTE — Social engineering during intake**
> Attackers also contact the SOC. Never ask for, or give, a password or one-time PIN (OTP). Verify phone, SMS and chat reporters by calling back the number in the company directory, not the number that contacted you. Record only the personal data the SOP needs (Data Privacy Act, RA 10173).

4. Red Flag Detections (PC 1.3)

The Competency Standards name two red flag detections that must be checked at once: ransomware and PE infection (virus). Ransomware encrypts files and demands payment; it can stop a bank branch, school or hospital within minutes. A PE infection is a file-infector virus: it inserts its code into Portable Executable files (.exe, .dll), so every copy of an infected installer spreads it further.

Code	Red Flag	What You See	Where
R1	Ransomware	Many files renamed with the same new extension in a short time	EDR file events, file shares
R2	Ransomware	A ransom note file appears (README_RESTORE_FILES.txt, HOW_TO_DECRYPT.txt)	Desktop, Documents, shares
R3	Ransomware	Shadow copy or backup deletion attempt (vssadmin delete shadows, wbadmin delete catalog)	EDR process events

Code	Red Flag	What You See	Where
R4	Ransomware	A detection name that contains "Ransom"	AV / EDR alerts
V1	PE infection	Many .exe or .dll files modified that should not change, by a non-updater process	EDR file events
V2	PE infection	A detection name that starts with "Virus:" (file infector)	AV alerts
V3	PE infection	The same detection on several hosts that used the same shared folder	AV alerts, SIEM

Look-alikes That Are NOT Red Flags

Some approved software does similar things: a backup agent creates thousands of archive files, and Windows Update replaces DLLs in System32. Check the process name and its signer (the company that digitally signed the program) against the SOP approved list. In the lab: scbackup.exe signed by SafeCopy Software Inc., and TiWorker.exe signed by Microsoft Windows. An unsigned program in a Temp or Downloads folder doing the same thing is a red flag.

> **TIP — First response to suspected ransomware (Level I)**
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> Ask the user to unplug the network cable or turn off Wi-Fi. Do NOT shut the PC down: evidence in memory is lost. Never pay or contact the attacker. Escalate immediately to the Shift Lead by phone, then record it. The Shift Lead decides on isolation.

5. Security Solution Scans and Actions (RK 1.7)

Antivirus and EDR tools scan in different ways and take different actions. Knowing them helps you read an alert correctly.

Scan or Action	Meaning
Real-time (on-access) protection	Checks each file when it is opened, downloaded or run
Quick scan	Checks the places malware usually hides (memory, startup, system folders)
Full scan	Checks every file on every drive; slow, used after an infection
Actions (CS Range of Variables)	Failed, Clean, Delete, Quarantine, Blocked, Re-image

Key Terms

Term	Meaning
SOP	Standard Operating Procedure: the written, approved steps every analyst follows
Acknowledge	Take ownership of an alert so everyone knows it is being handled

Term	Meaning
SLA breach	Missing the time limit the SOP sets for a severity
Detection report	A report of a possible incident from a person (phone, walk-in, email, SMS, chat, video conference)
Red flag detection	A detection the SOP says must be escalated at once (ransomware, PE infection)
PE file	Portable Executable: a Windows program or library (.exe, .dll)
Signer	The company that digitally signed a program; unsigned programs deserve a closer look
Social engineering	Tricking people, including SOC staff, into giving access or information

References

- TESDA Competency Standards — Cyber Threat Monitoring Level I (ICT Sector), unit CS-ICT251101, Element 1 and Range of Variables.
- NIST SP 800-61 Rev. 3, Incident Response Recommendations and Considerations for Cybersecurity Risk Management.
- NIST SP 800-83 Rev. 1, Guide to Malware Incident Prevention and Handling for Desktops and Laptops.
- CISA #StopRansomware Guide (cisa.gov/stopransomware).
- MITRE ATT&CK® T1486 Data Encrypted for Impact and T1490 Inhibit System Recovery — attack.mitre.org.
- Republic Act No. 10175 (Cybercrime Prevention Act of 2012) and Republic Act No. 10173 (Data Privacy Act of 2012).